

## Q Longest Increasing Subsequence (LIS)

Given an array. Return the length of LIS.

Ex 10, 22, 9, 33, 21, 50, 41, 60, 80  $\rightarrow$  6

10, 33, 60  $\rightarrow$  3

9, 21, 50  $\rightarrow$  3

9, 21, 41, 60  $\rightarrow$  4

10, 22, 33, 50, 60, 80  $\rightarrow$  6

### Brute Force

- Iterate over all sub sequences  $\rightarrow 2^N$
- Check if they are increasing  $\rightarrow N$   
(maintain the max length)

TC:  $O(N \cdot 2^N)$

5, 8, 10, 6, 9, 11

5

5, 8

5, 8, 10

5, 6

5, 6, 9

5, 8, 10, 11

A: 5, 8, 10, 6, 9, 11, 7  
 1 2 3 2 3

| A[i] | Length of LIS<br>ending at A[i] | sub seq.                            |
|------|---------------------------------|-------------------------------------|
| 5    | 1                               | {5}                                 |
| 8    | 2                               | {5, 8}                              |
| 10   | 3                               | {5, 8, 10}                          |
| 6    | 2                               | {5, 6}                              |
| 9    | 3                               | {5, 6, 9} / {5, 8, 9}               |
| 11   | 4                               | {5, 6, 9, 11}<br>{5, 8, 9, 11}<br>⋮ |
| 7    | 3                               | {5, 6, 7}<br>⋮                      |

TC:  $O(N^2)$

SC:  $O(N)$



$O(N \log N)$

(Binary Search)

(Free Time)

{ Ass done (> 90%)  
 HW done (> 90%)  
 Revision done (DSA)

## Q. Russian doll envelope

Given heights & widths of  $N$  envelope.

Create a seq by putting envelopes inside one another.

You can put  $E_1$  inside  $E_2$

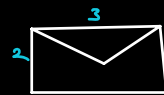
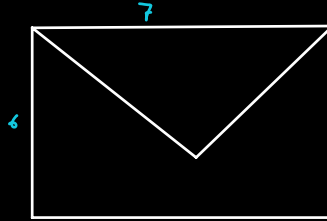
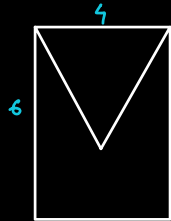
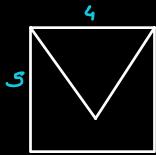
iff  $H_1 < H_2$

$W_1 < W_2$

Rotating & folding is not allowed.

$H : 5 \quad 6 \quad 6 \quad 2$

$W : 4 \quad 4 \quad 7 \quad 3$



$$\begin{matrix} 2 & & 5 & & 6 \\ 3 & < & 4 & < & 7 \end{matrix} \rightarrow 3$$

$(2, 2)$

$(3, 4)$

$(5, 7)$

$(10, 13)$

$(18, 20)$

$$h_1 < h_2 < h_3 < h_4 < h_5$$

$$w_1 < w_2 < w_3 < w_4 < w_5$$

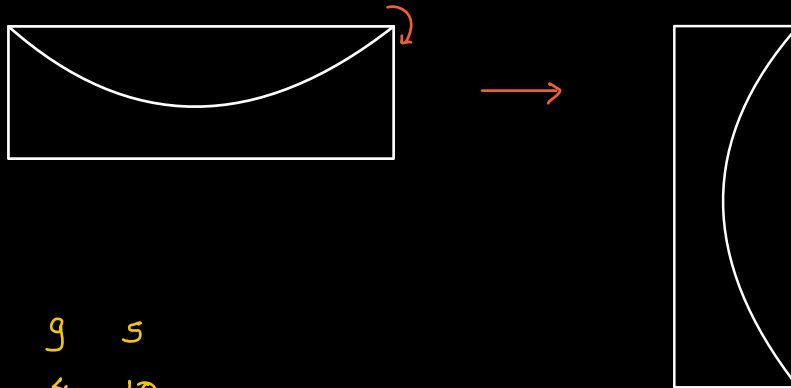
|       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|
| $h_1$ | $h_2$ | $h_3$ | $h_4$ | $h_5$ | $h_6$ |
| $w_1$ | $w_2$ | $w_3$ | $w_4$ | $w_5$ | $w_6$ |

|      |   |   |   |   |    |    |
|------|---|---|---|---|----|----|
| 4    | 2 | 5 | 7 | 8 | 12 | 15 |
| W    | 7 | 1 | 3 | 2 | 6  | 8  |
| LIS: | 1 | 1 | 2 | 2 | 3  | 4  |

TC:  $O(N \log N)$  +  $O(N^2)$   $\longrightarrow$   $O(N^2)$   
                     Sorting                      LIS

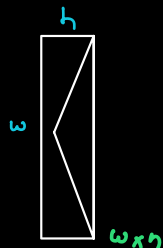
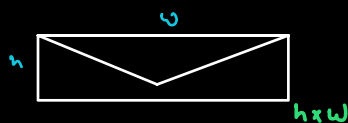
Sc:  $O(1)$

Q In the previous problem rotation is allowed.



|    |   |   |    |
|----|---|---|----|
| h: | 1 | 9 | 5  |
| w: | 3 | 4 | 10 |

$\frac{1}{3} < \frac{9}{4} < \frac{10}{5} \longrightarrow 3$



|   |   |                     |
|---|---|---------------------|
| A | B |                     |
| h | w | $\longrightarrow 1$ |
| w | h |                     |

h: 1 9 5 3 4 10  
 w: 3 4 10 1 9 5

sort w[]  
 h 3 1 9 10 4 5  
 w 1 3 4 5 9 10

LIS in h[]  
 3

Google

0/1 Knapsack

Weight limit  $K \leq 100$

$0 \leq V[i] \leq 100$

$N \leq 100$

TC =  $O(NK)$   
 (old)

$f_n(n, K) \rightarrow$  max value which can be created  
 using the first  $n$  items under  
 wt limit of  $K$ .

TC:  $O(NK)$

Hint

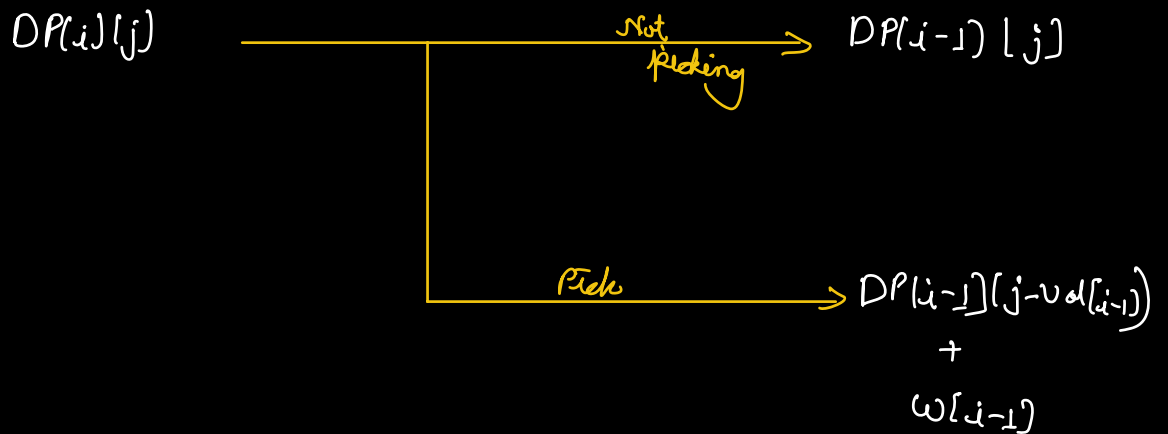
Come up with a new state definition which does  
 not have  $K$  in fn calls.

$$\text{Max possible value} = \sum_{i=0}^N v[i]$$

(K)

$f(n, val) \rightarrow \text{wt}$

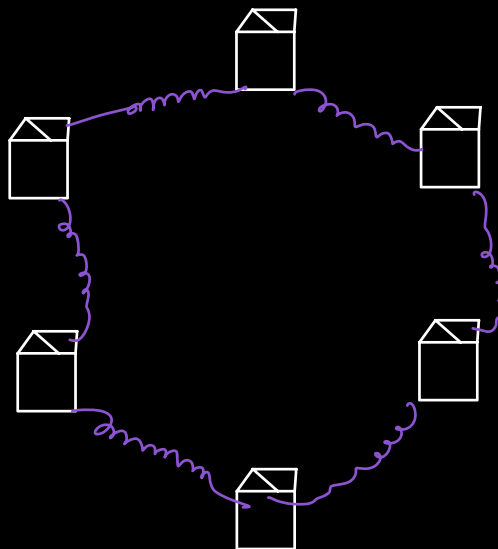
min weight required to get val  
using first  $n$  items.



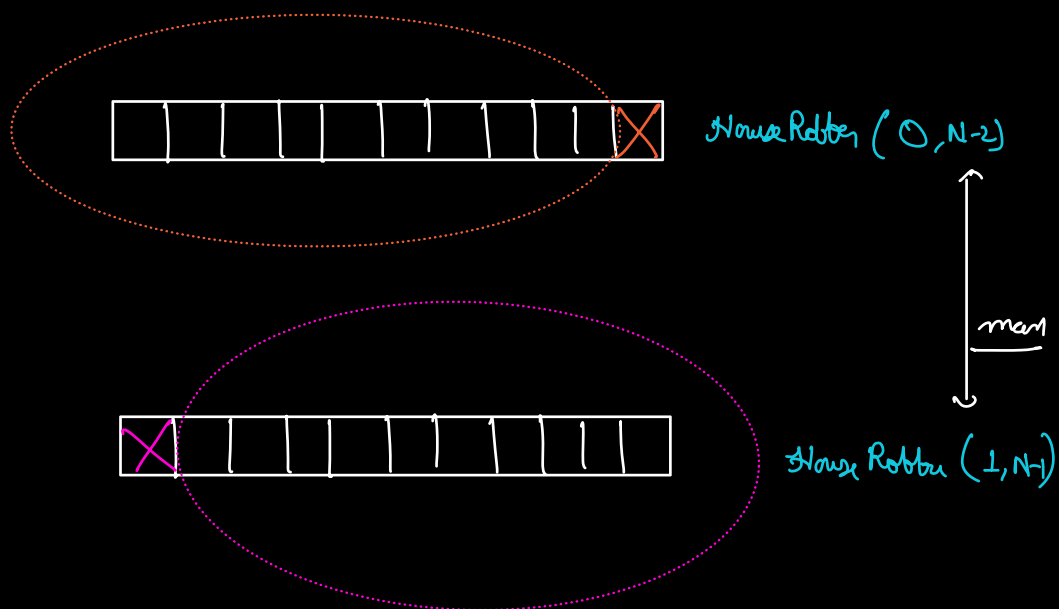
## Q House Robber II

Ex 2, 3, 2  $\rightarrow 3$

Ex 1, 2, 3, 1  $\rightarrow 4$



Observation: Only one of last & first will be present in the answer.



## Revision

- Level order traversal (Tree)
- BFS (Queues)